LAF SYSTEM

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Task 01: Project Definition

1. Project Title:

- Lost and Found System **(LAF SYSTEM).**

2. Problem Statement:

- Usually in a crowded area it's very easy to lose a lot of important things. That has happened to every one of us. Maybe, for example, car keys, maybe locker keys, maybe AirPods, or money. These are simple everyday things we can lose.

3. Objectives:

- **O1:** Let anyone sign up using their national ID.
- **O2:** Describe what they precisely lost.
- **O3:** Add proof that what is found belongs to them.

4. Scope:

- **IN:** Registration, reporting lost and found items, searching, contacting the founder and who lost it, and making reviews.
- **OUT:** Payments, delivery system, live location.

5. Stakeholders:

- The founder, the one who lost it, app security managers, the software engineers, and software developers.

Task 02: Elicitation plan and role-play

1. Techniques:

- **Interviews**: to check for feedback.
- **Ethnography:** to check how they get what they lost from the founders.
- **Stories and Scenarios:** to check the story of those who have lost the item for enough evidence.

2. Interview Questions:

- **Open Ended Questions:**
    - What do you face when you lose an item?
    - How do you search for what you lost?
    - What process would you suggest for the current lose-and-found process?
- **Close Ended Questions:**
    - Have you ever lost an item? (Yes/No)
    - Have you ever found an item that belongs to another student? (Yes/No)
    - Would this system make finding lost items easier to be found? (Yes/No)

3. **Role play:**

Stakeholder	Technique	Questions or observations	Needs discovered
Teacher	Interview	Open Ended and Close Ended Questions	Nothing. It validated our work.

Task 03: Write user stories and acceptance criteria

1. **8 Functional Stories and 5 Non-Functional Requirements:**

Stories with acceptance criteria:

- **US-01 (Must), Register.** As a stakeholder, I want to log in and have my own account so that I can log in.
    - **AC1:** Given I log in, when I forget my password, then I should be able to reset it.
    - **AC2:** Given that I can't reset the password, when I enter my details, then customer services should be able to assist.
- **US-02 (Must), Reporting lost items.** As a stakeholder, I want to be able to post so that it's easier for reachability.
    - **AC1:** Given that I find a lost item, when I find it, then I should post it.
    - **AC2:** Given that I have lost an item, when I search for it, I should be able to filter it.
    - **AC3:** Given that I found what I lost, when I find it, I should be able to identify it.
    - **AC4:** Given that I found what someone else lost, when I find the owner, then I should make sure it belongs to them.
- **US-04 (Should): ID Tags.** As a stakeholder, I want what I have lost to be found to have ID tags. So that I can return it.
    - **AC1:** Given that I found an item, when I post it, then the app should auto-generate a tag for that item.
    - **AC2:** Given that I found my lost item, when I open the post, then I should see the ID tag.

Non-functional requirements:

ID	Requirement	Type
NFR-01	Loading shall appear within 0.5-2 seconds for 90% of the time.	**Product:** Performance
NFR-02	User experience shall be very simple And easy for 90% of our users.	**Product:** Usability
NFR-03	The user interface shall be basic, understandable, and well organized.	Organizational
NFR-04	The system shall store the student IDs as hexadecimal for security reasons.	**External:** Legislative (Privacy)
NFR-05	The system should 99% of the time be available.	**Product:** Reliability

Task 04: Classify

- **Conflicts: NFR-1 & NFR-04**
  - **NFR-01:** The system should load within 0.5-2 seconds.
  - **NFR-04:** The system shall store the student IDs as hexadecimal for security reasons.

- **How to resolve:**
  - Hexadecimal alone doesn't provide security; secure hashing would be needed depending on how the student IDs are used.

Task 05: Prioritize, build the backlog and trace

1. Product Backlog table:

ID	Item	Type	Priority	Estimate (points)	Source
US-01	Register	Functional	Must	8	Stakeholders interview
US-02	Reporting lost items	Functional	Must	5	Stakeholders interview
US-03	ID Tags	Functional	Should	3	Stakeholders interview
NFR-03	UI	Organizational	Must	5	Developers
NFR-01	Search Performance	Product	should	3	Developers

2. Traceability table:

Stakeholder need	Requirement(s)	Acceptance tests
N1: Users need to login	US-01, NFR-01, NFR-03	US-01 AC1
N2: Users must report lost items	NFR-02	US-04 AC1-AC2 -AC3- AC4
N3: Customer service must always be available for user	NFR-05	US-01 AC2